

Title

Coding and Solution Implementation – SB Stocks

Introduction

SB Stocks was implemented as a MERN stack web application.

The application combines a React frontend, Express backend, and MongoDB database.

User Authentication Implementation

User registration validates account information before storing user data.

Passwords are hashed before database storage.

During login, the entered password is verified against the stored password hash.

After successful authentication, a JWT is generated.

Protected APIs validate the JWT before processing user requests.

Stock Module Implementation

The stock module provides:

- Stock listings.
- Stock search.
- Stock details.
- Latest available prices.
- Historical stock data.

Stock information is processed by backend APIs and displayed through React components.

Portfolio Implementation

Each portfolio is associated with a user.

A user can create multiple portfolios.

Portfolio holdings maintain:

- Stock symbol.
- Quantity.

- Average buy price.

Portfolio values are calculated using stock prices.

Buy Implementation

When a user purchases a stock:

1. The user is authenticated.
2. Quantity is validated.
3. Stock information is retrieved.
4. Total purchase value is calculated.
5. Virtual balance is verified.
6. Portfolio ownership is verified.
7. Holdings are updated.
8. Virtual balance is reduced.
9. A transaction is recorded.
10. An order is recorded.

Sell Implementation

When a user sells a stock:

1. User authentication is verified.
2. Portfolio ownership is checked.
3. Existing holdings are verified.
4. Available quantity is checked.
5. Sale value is calculated.
6. Holdings are reduced.
7. Virtual balance is increased.
8. A transaction is recorded.
9. An order is recorded.

Strategy Analysis

Historical stock data is analyzed using a basic moving average strategy.

Short-term and long-term moving averages are calculated.

Trading signals are identified based on moving average crossover conditions.

The results are presented for educational purposes.

Admin Module

The administrative module provides access to:

- System statistics.
- Registered users.
- Orders.
- Transactions.
- Stock data.

Admin routes use authorization checks.

Conclusion

The coding solution implements the required virtual stock trading workflow while maintaining modularity and secure backend validation.

☐ Number of Functional Features Included in the Solution

Title

Functional Features Included in SB Stocks

The SB Stocks solution includes the following functional features:

1. User Registration

Allows users to create accounts.

2. User Login

Provides secure account authentication.

3. JWT Authentication

Protects private application APIs.

4. Personalized Dashboard

Displays account and trading information.

5. Virtual Balance

Provides simulated trading funds.

6. Stock Browsing

Displays available stocks.

7. Stock Search

Searches stocks by symbol or company.

8. Stock Details

Displays detailed stock information.

9. Historical Stock Data

Provides historical stock prices.

10. Stock Charts

Visualizes historical stock trends.

11. Multiple Virtual Portfolios

Allows users to create multiple portfolios.

12. Portfolio Holdings

Displays stocks owned by users.

13. Virtual Stock Buying

Simulates stock purchases.

14. Virtual Stock Selling

Simulates stock sales.

15. Balance Validation

Prevents purchases with insufficient virtual funds.

16. Holdings Validation

Prevents users from selling unavailable stocks.

17. Quantity Validation

Prevents invalid trading quantities.

18. Transaction Recording

Records BUY and SELL activities.

19. Order Recording

Maintains stock order information.

20. Transaction History

Displays previous trading activities.

21. Portfolio Value Calculation

Calculates current portfolio value.

22. Profit/Loss Calculation

Calculates portfolio performance.

23. Average Buy Price Calculation

Maintains weighted average purchase price.

24. Strategy Analysis

Analyzes historical stock data.

25. Profile Management

Displays and manages permitted profile information.

26. Admin Dashboard

Displays system statistics.

27. User Monitoring

Allows admins to view users.

28. Order Monitoring

Allows admins to view orders.

29. Transaction Monitoring

Allows admins to view transactions.

30. Admin Stock Analysis

Allows admins to inspect stock charts.

31. Protected Routes

Restricts pages based on authentication.

32. Role-Based Authorization

Protects admin functionality.

33. Responsive Navigation

Provides role-based navigation.

34. Secure Logout

Ends the authenticated session.

Total Functional Features

34 major functional features are included in the SB Stocks solution.